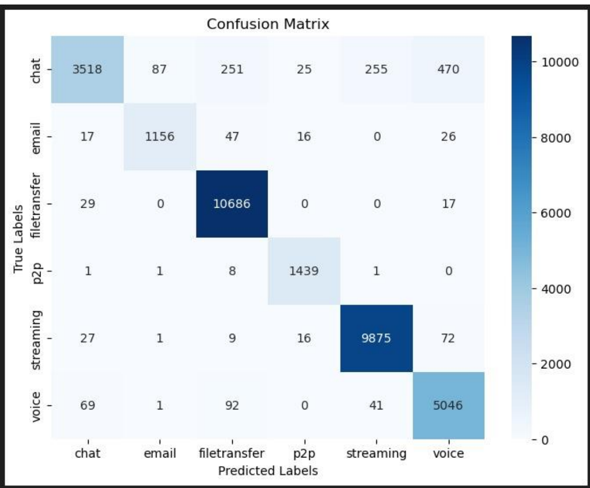
INSTALACION GPU ALL READY: https://www.tensorflow.org/install/pip?hl=es-419#windows-native

RESULTADOS CON DATA LEAKING:

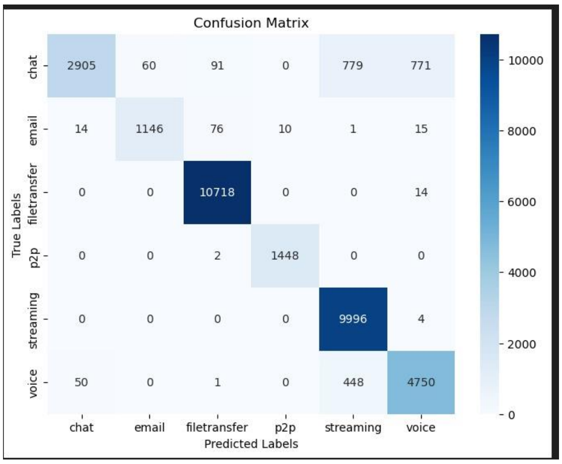
Normal data



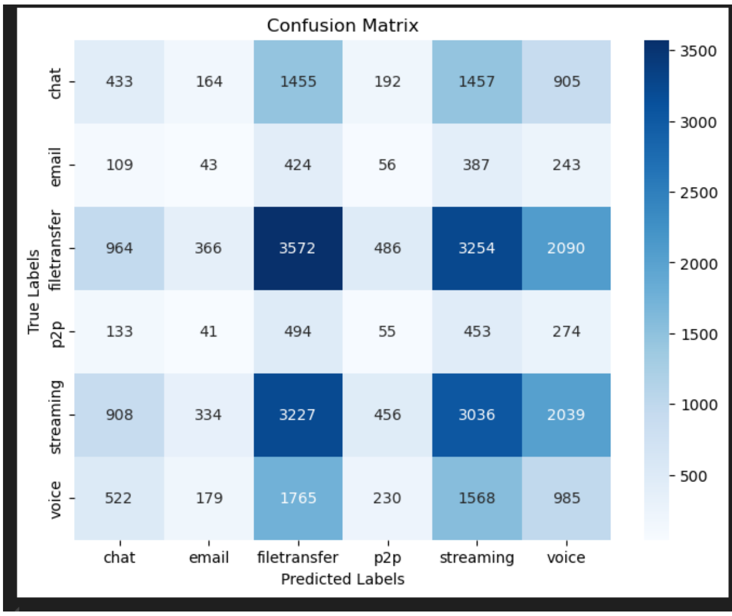
Classification Report:

	precision	recall	f1-score	support
chat	0.96	0.76	0.85	4606
email	0.93	0.92	0.92	1262
filetransfer	0.96	1.00	0.98	10732
p2p	0.96	0.99	0.98	1450
streaming	0.97	0.99	0.98	10000
voice	0.90	0.96	0.93	5249
accuracy			0.95	33299
macro avg	0.95	0.94	0.94	33299
weighted avg	0.95	0.95	0.95	33299

Morton data



Una vez eliminado el data leak, estos son los resultados para Morton:

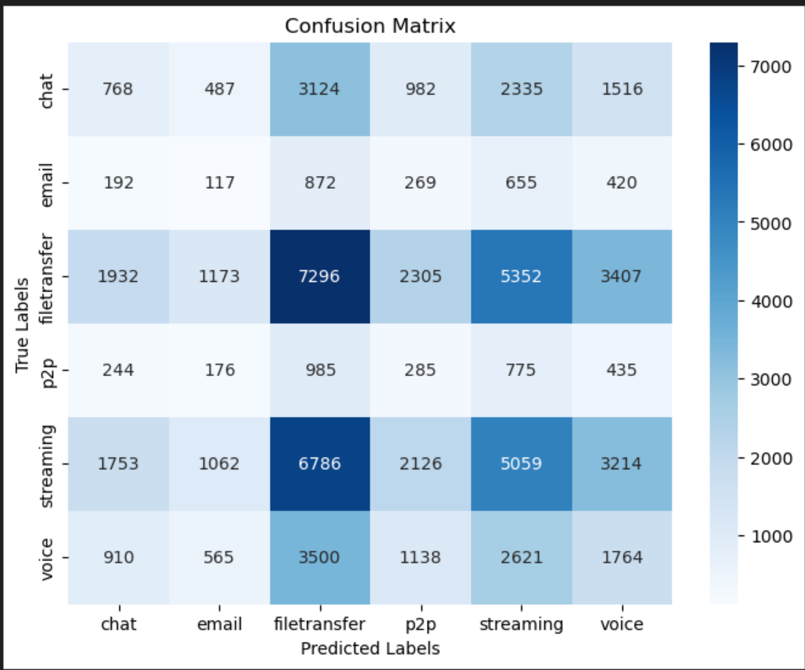


Classification Report:

	precision	recall	f1-score	support
chat	0.14	0.09	0.11	4606
email	0.04	0.03	0.04	1262
filetransfer	0.33	0.33	0.33	10732
p2p	0.04	0.04	0.04	1450
streaming	0.30	0.30	0.30	10000
voice	0.15	0.19	0.17	5249
accuracy			0.24	33299
macro avg	0.17	0.17	0.16	33299
weighted avg	0.24	0.24	0.24	33299

CON DATA AUGMENTATION:

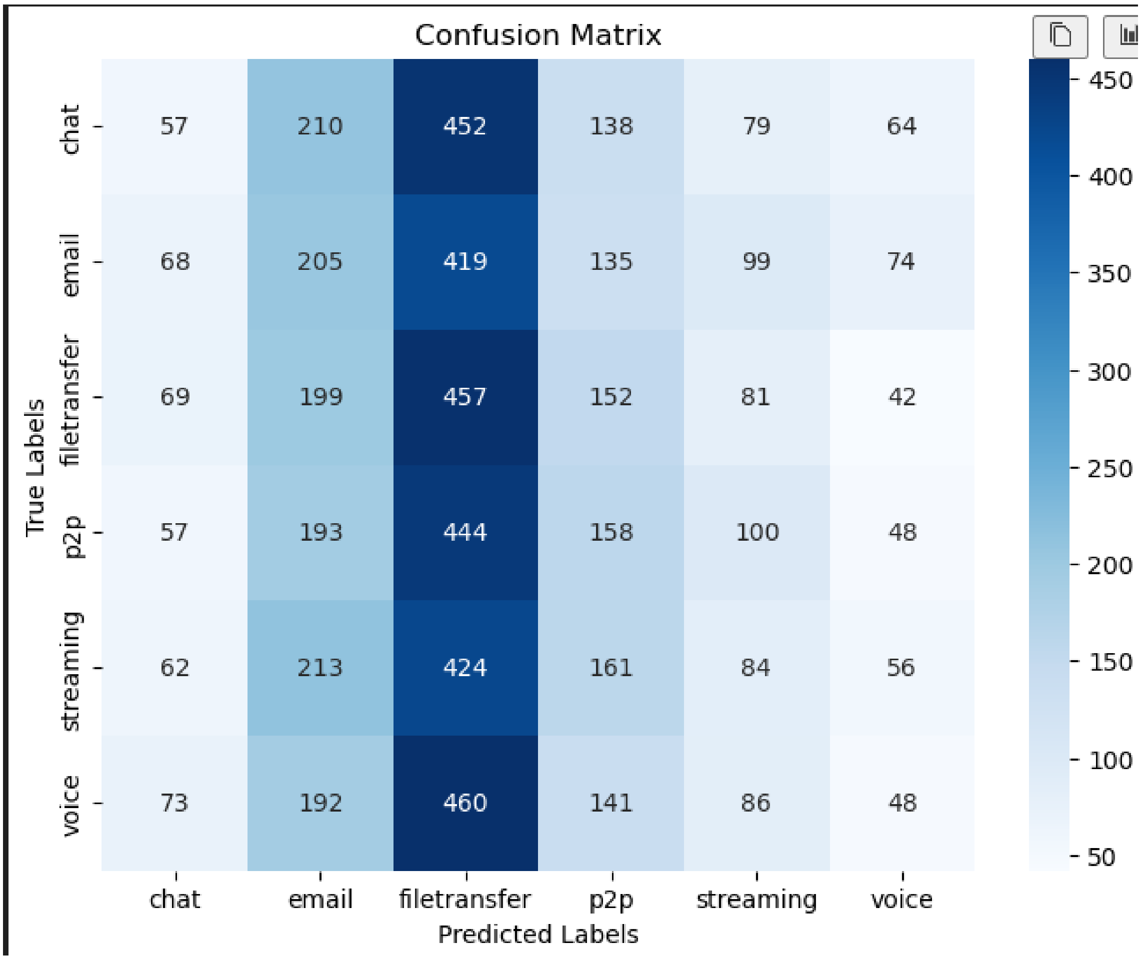
```
3747/3747 [==============================] - 1464s 390ms/step - loss: 1.1513 - accuracy: 0.5349
Epoch 2/3
3747/3747 [==============================] - 1471s 393ms/step - loss: 0.9883 - accuracy: 0.6052
Epoch 3/3
3747/3747 [==============================] - 1483s 396ms/step - loss: 0.9019 - accuracy: 0.6438
1041/1041 [==============================] - 24s 23ms/step
```




```
📊 Classification Report:

              precision    recall  f1-score   support

        chat       0.13      0.08      0.10      9212
       email       0.03      0.05      0.04      2525
filetransfer       0.32      0.34      0.33     21465
         p2p       0.04      0.10      0.06      2900
   streaming       0.30      0.25      0.27     20000
       voice       0.16      0.17      0.17     10498

    accuracy                           0.23     66600
   macro avg       0.17      0.16      0.16     66600
weighted avg       0.24      0.23      0.23     66600
```

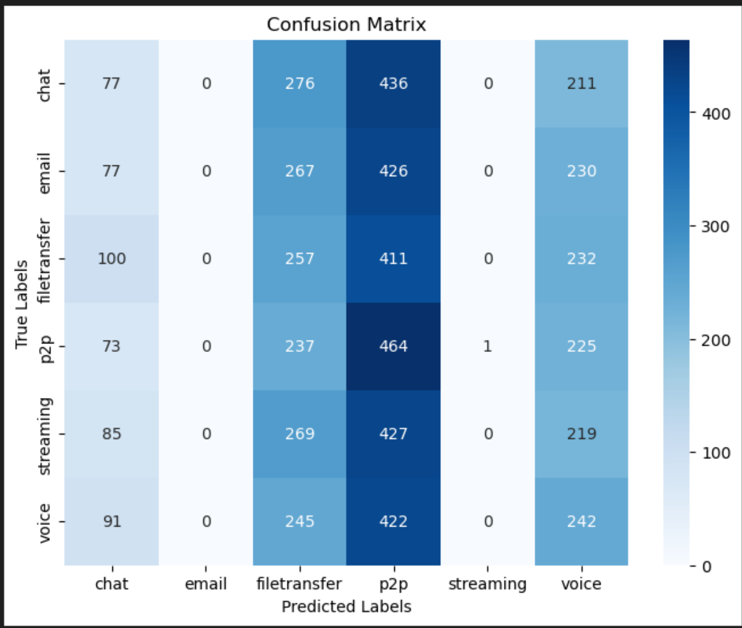
En resumen, hay que hacer a mano la igualación en el numero de imágenes para todas las clases, si no, seguirá habiendo huge bias hacia streaming y file transfer

FIXEADO A 10K el numero de datos para cada clase:



Ahora lo que pasa es que esta turbo biased hacia filetransfer y email, no es capaz la red neuronal básica (modeloV2) de entender una mierda

```
Epoch 1/3
844/844 [==============================] - 442s 508ms/step - loss: 1.5175 - accuracy: 0.3141 - val_loss: 1.5056 -
Epoch 2/3
844/844 [==============================] - 442s 523ms/step - loss: 1.4784 - accuracy: 0.3325 - val_loss: 1.5281 -
Epoch 3/3
844/844 [==============================] - 392s 464ms/step - loss: 1.4648 - accuracy: 0.3389 - val_loss: 1.5722 -
94/94 [==============================] - 30s 304ms/step
```



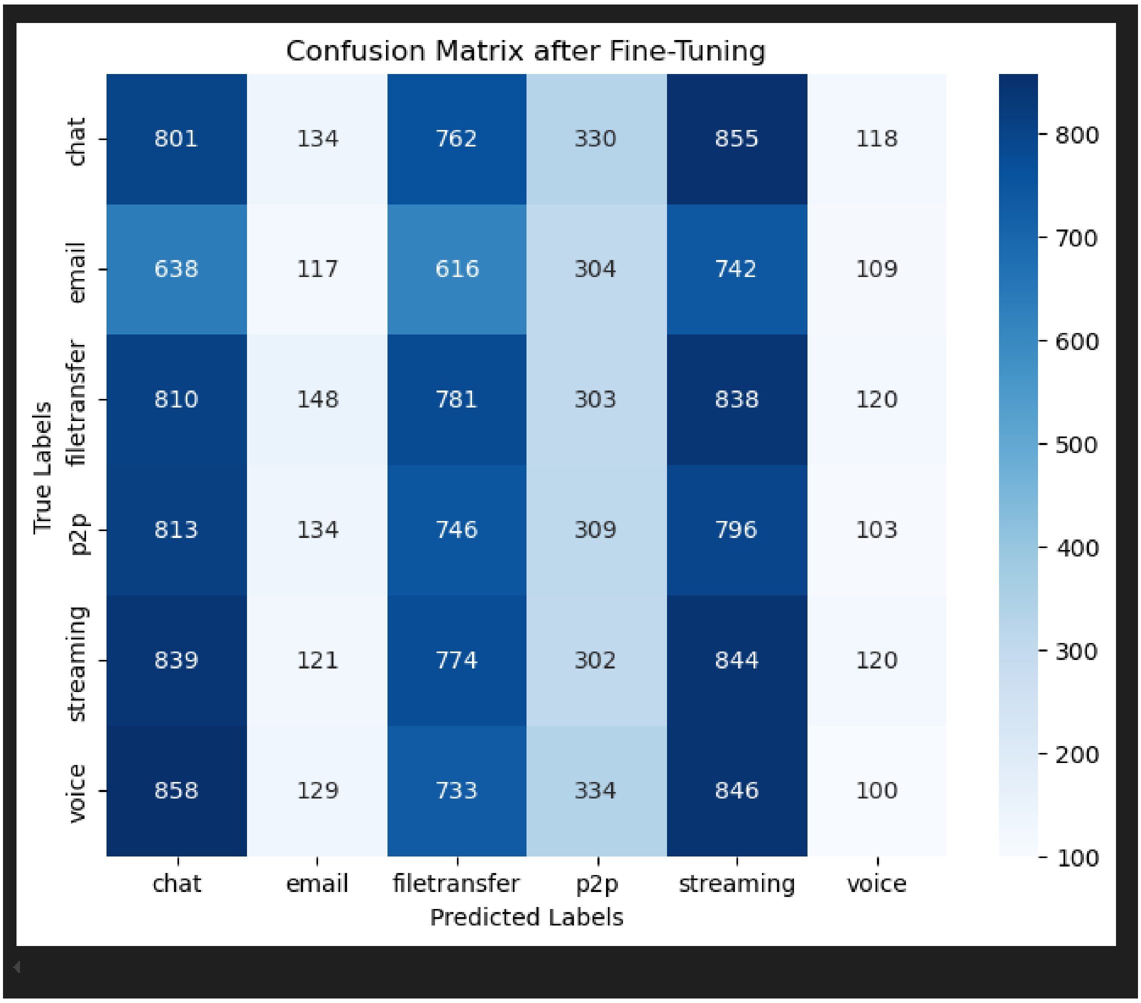
📊 Classification Report:

	precision	recall	f1-score	support
chat	0.15	0.08	0.10	1000
email	0.00	0.00	0.00	1000
filetransfer	0.17	0.26	0.20	1000
p2p	0.18	0.46	0.26	1000
streaming	0.00	0.00	0.00	1000
voice	0.18	0.24	0.21	1000
accuracy			0.17	6000
macro avg	0.11	0.17	0.13	6000
weighted avg	0.11	0.17	0.13	6000

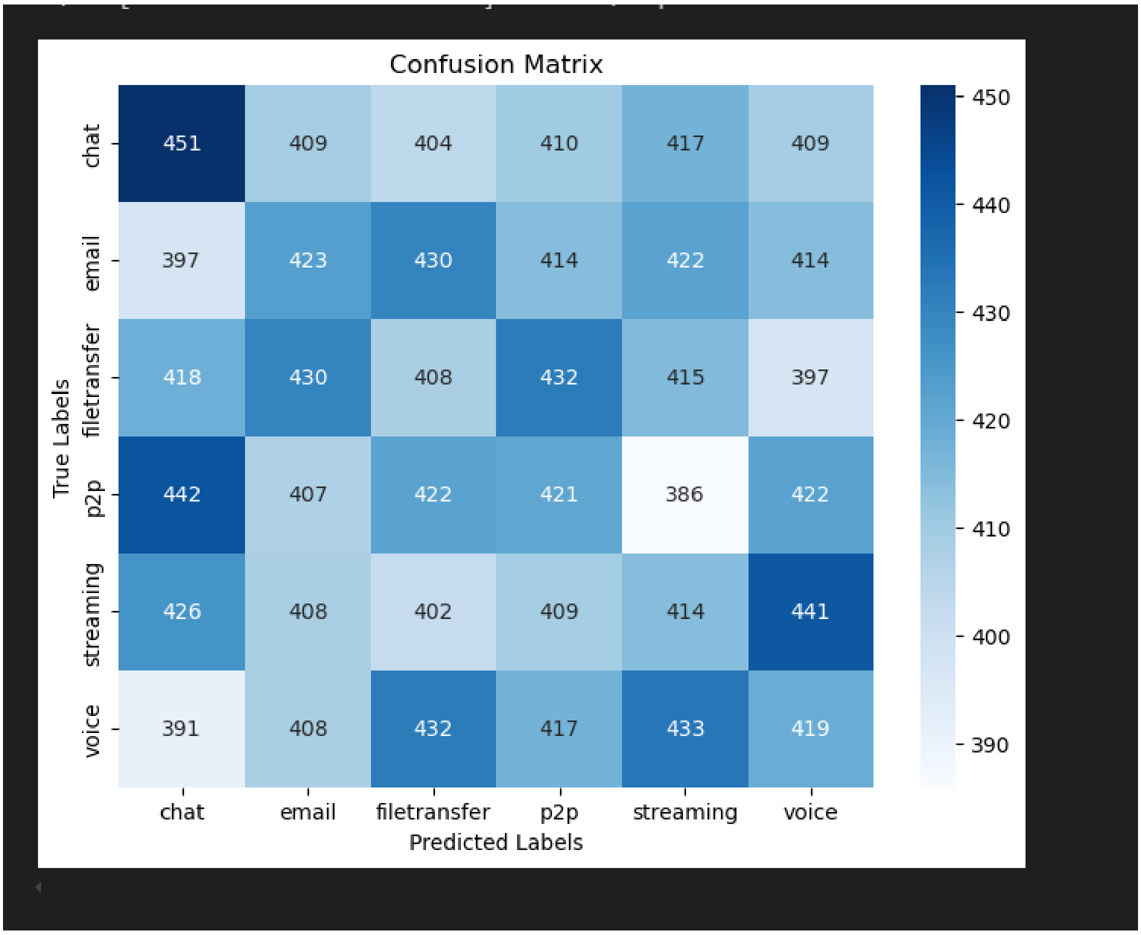
RESNET suposedly

```
Epoch 1/5
844/844 [==============================] - 430s 504ms/step - loss: 2.0355 - accuracy: 0.3379
Epoch 2/5
844/844 [==============================] - 413s 489ms/step - loss: 1.4146 - accuracy: 0.3623
Epoch 3/5
844/844 [==============================] - 411s 487ms/step - loss: 1.3958 - accuracy: 0.3756
Epoch 4/5
844/844 [==============================] - 410s 486ms/step - loss: 1.3784 - accuracy: 0.3899
Epoch 5/5
844/844 [==============================] - 415s 492ms/step - loss: 1.3615 - accuracy: 0.4031
94/94 [==============================] - 28s 299ms/step - loss: 1.3993 - accuracy: 0.3958
Test Accuracy after fine-tuning: 39.58%
```

RESNET con datos normales:



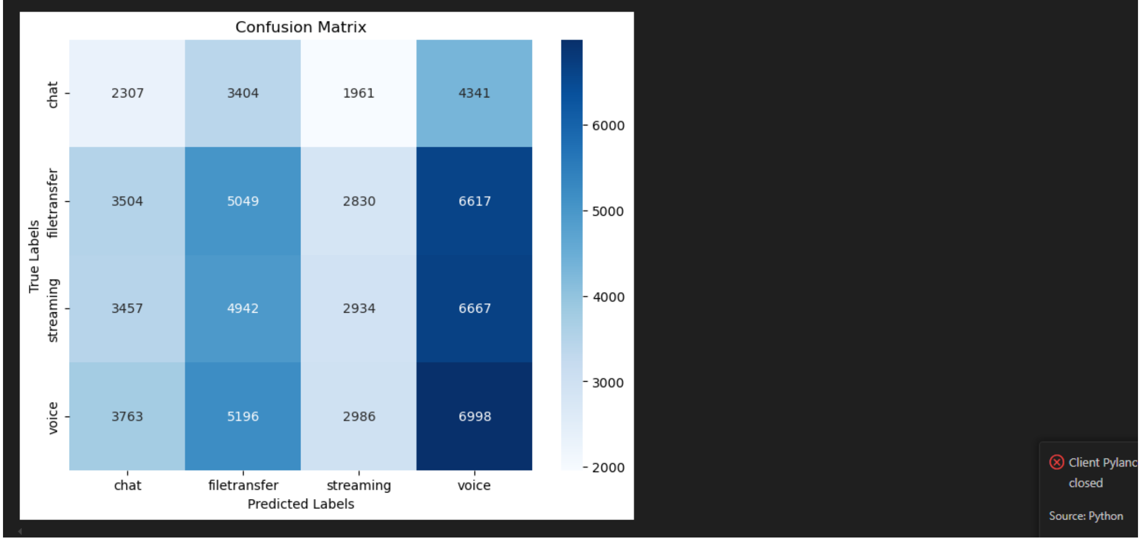
VOLVIENDO A MODELO NUESTRO con datos normales:




```
Epoch 6/6
938/938 [==============================] - 28s 30ms/step - loss: 0.0769 - accuracy: 0.9731 - val_loss: 0.0346 - val_accuracy: 0.9865 - lr: 0.0010
235/235 [==============================] - 5s 22ms/step
```

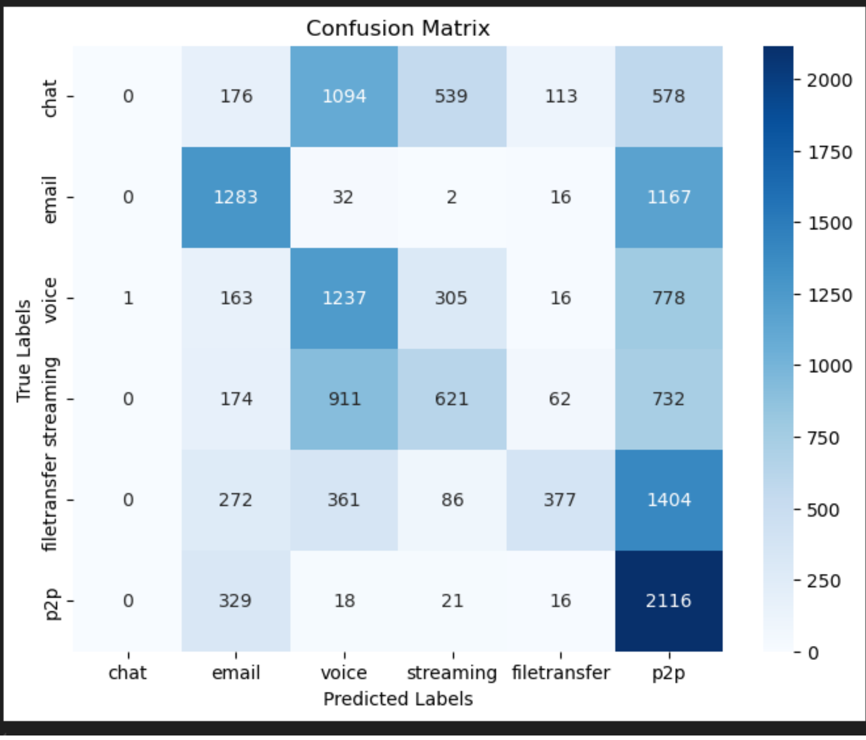
MORTON CON TODOS LOS DATOS.

```
Found 379422 images belonging to 4 classes.
Found 66956 images belonging to 4 classes.
Epoch 1/5
5929/5929 [==============================] - 3216s 542ms/step - loss: 0.3576 - accuracy: 0.8560 - val_loss: 2.2832 - val_accuracy: 0.6064 - lr: 0.0010
Epoch 2/5
5929/5929 [==============================] - 948s 160ms/step - loss: 0.0965 - accuracy: 0.9701 - val_loss: 2.0130 - val_accuracy: 0.6154 - lr: 0.0010
Epoch 3/5
5929/5929 [==============================] - 449s 76ms/step - loss: 0.0678 - accuracy: 0.9788 - val_loss: 2.1901 - val_accuracy: 0.5860 - lr: 0.0010
Epoch 4/5
5929/5929 [==============================] - 377s 64ms/step - loss: 0.0554 - accuracy: 0.9823 - val_loss: 2.2422 - val_accuracy: 0.6275 - lr: 0.0010
Epoch 5/5
5929/5929 [==============================] - ETA: 0s - loss: 0.0471 - accuracy: 0.9846
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
5929/5929 [==============================] - 337s 57ms/step - loss: 0.0471 - accuracy: 0.9846 - val_loss: 2.5240 - val_accuracy: 0.6320 - lr: 0.0010
1047/1047 [==============================] - 27s 26ms/step
```

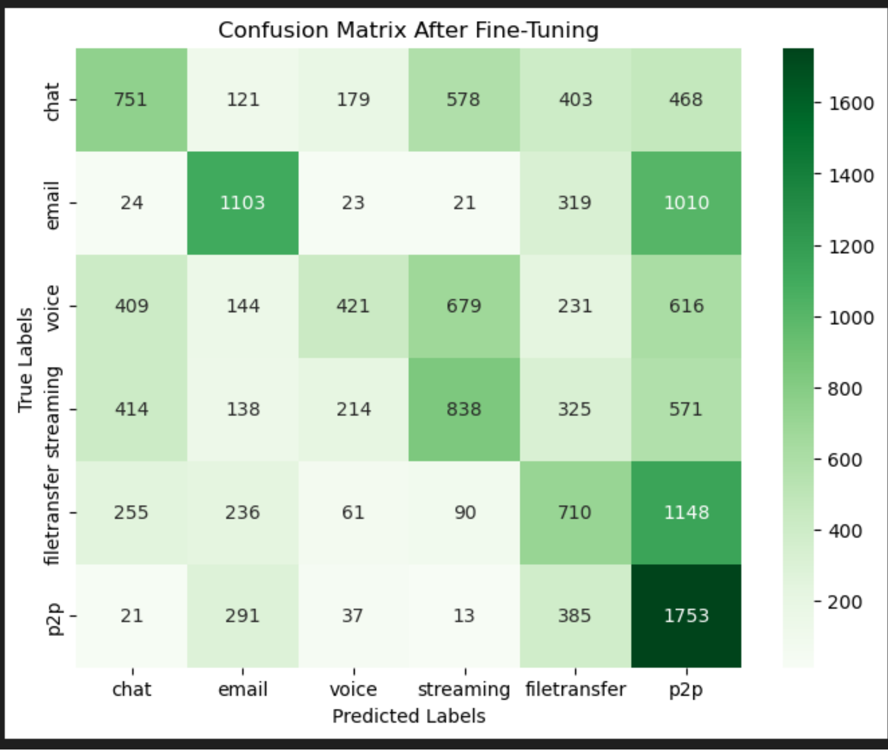


CON DATAFRAME NO CON IMAGEGENERATION: 12.5K NORMAL

```
Epoch 1/3
938/938 [==============================] - 29s 28ms/step - loss: 1.5736 - accuracy: 0.3208 - val_loss: 1.5158
Epoch 2/3
938/938 [==============================] - 25s 26ms/step - loss: 1.5035 - accuracy: 0.3548 - val_loss: 1.4858
Epoch 3/3
938/938 [==============================] - 26s 27ms/step - loss: 1.4728 - accuracy: 0.3753 - val_loss: 1.4696
235/235 [==============================] - 5s 19ms/step
```

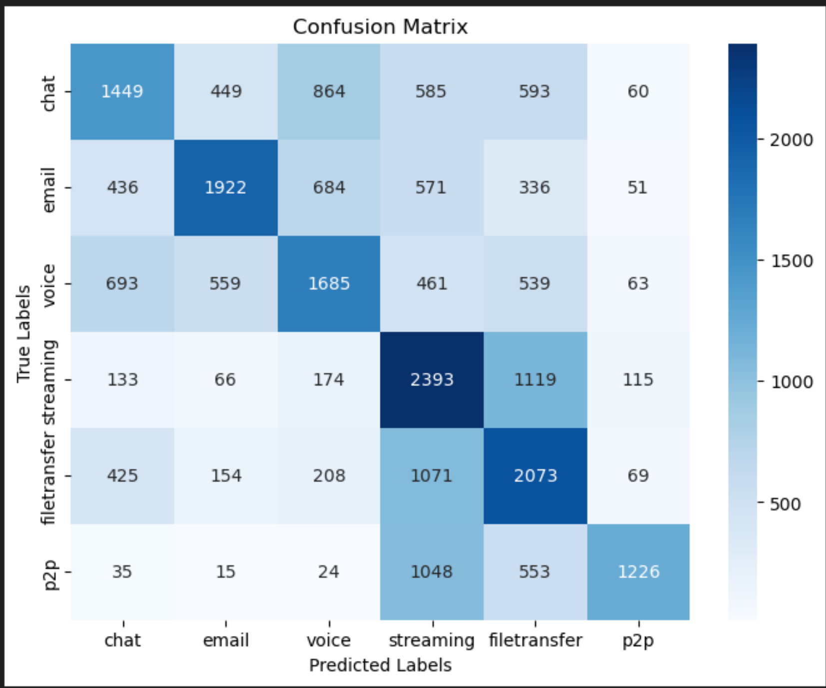


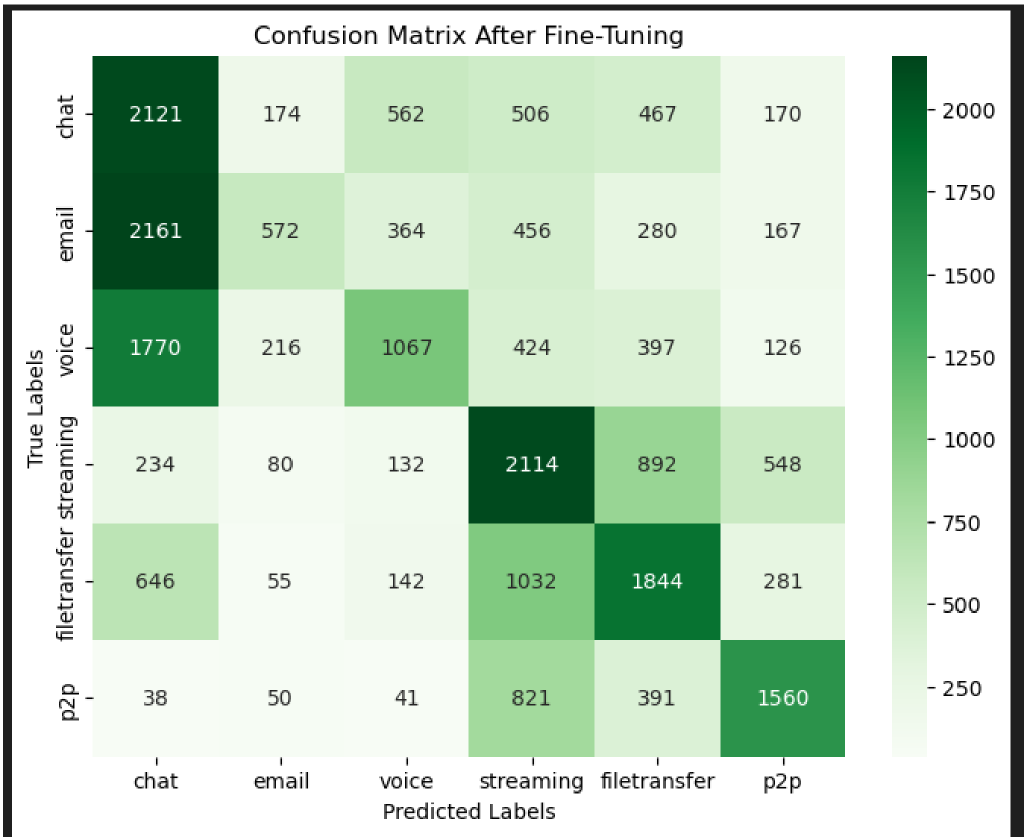

```
938/938 [==============================] - 51s 51ms/step - loss: 1.5776 - accuracy: 0.3424 - val_loss: 1.5365
235/235 [==============================] - 5s 20ms/step
```



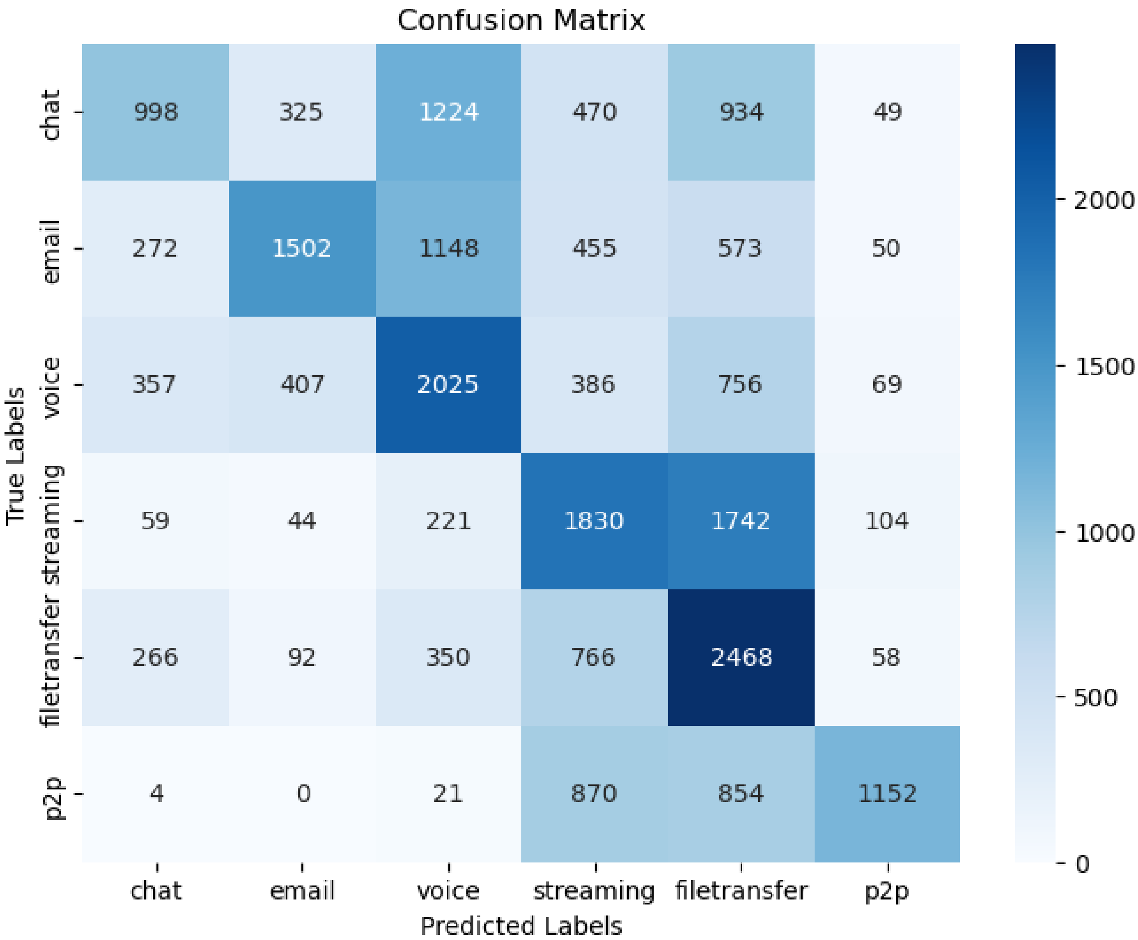
DATAFRAME MORTON20K 25 capas:

```
================] - 40s 28ms/step - loss: 1.3702 - accuracy: 0.4519 - val_loss: 1.3671 - val_accuracy: 0.4574

================] - 40s 28ms/step - loss: 1.3628 - accuracy: 0.4542 - val_loss: 1.3588 - val_accuracy: 0.4674

================] - 41s 28ms/step - loss: 1.3540 - accuracy: 0.4614 - val_loss: 1.3552 - val_accuracy: 0.4693
==============] - 8s 19ms/step
```

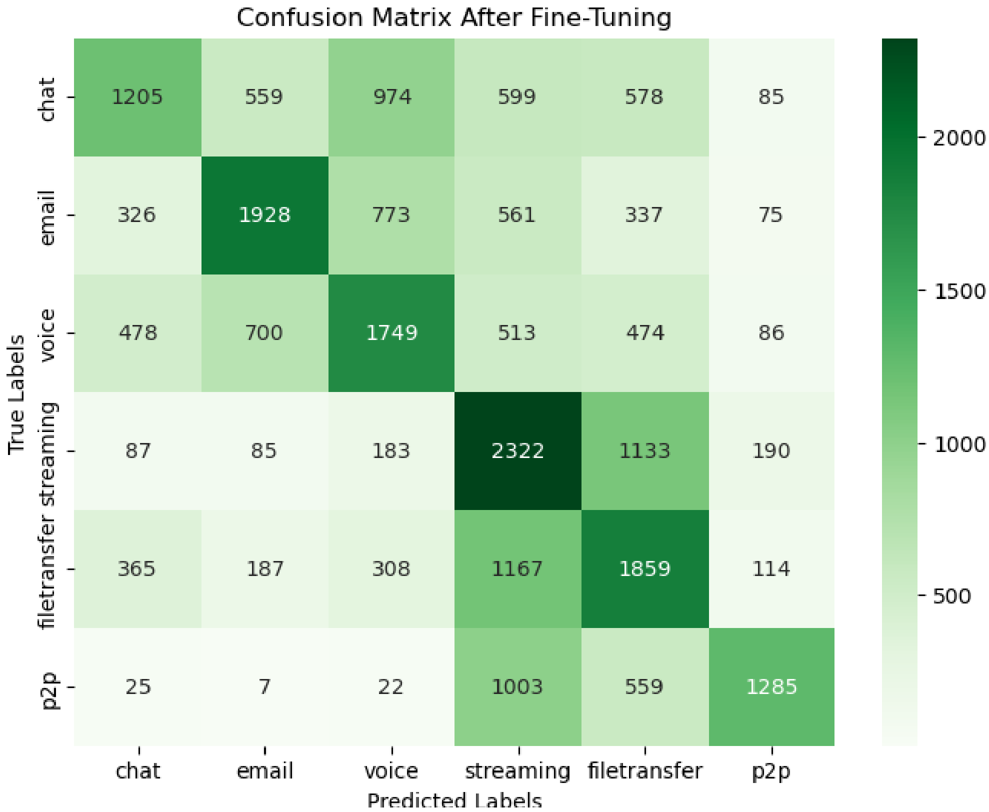




Otra run:



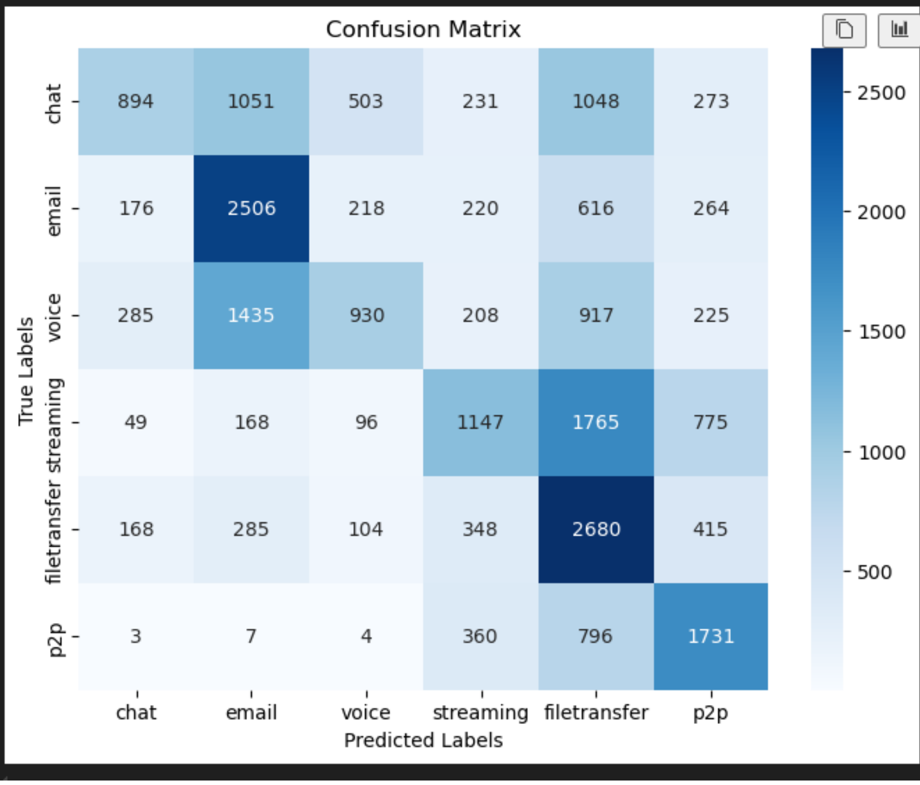
Confusion Matrix
True Labels
Predicted Labels
chat
email
voice
streaming
filetransfer
p2p
998 325 1224 470 934 49
272 1502 1148 455 573 50
357 407 2025 386 756 69
59 44 221 1830 1742 104
266 92 350 766 2468 58
4 0 21 870 854 1152
2000
1500
1000
500
0



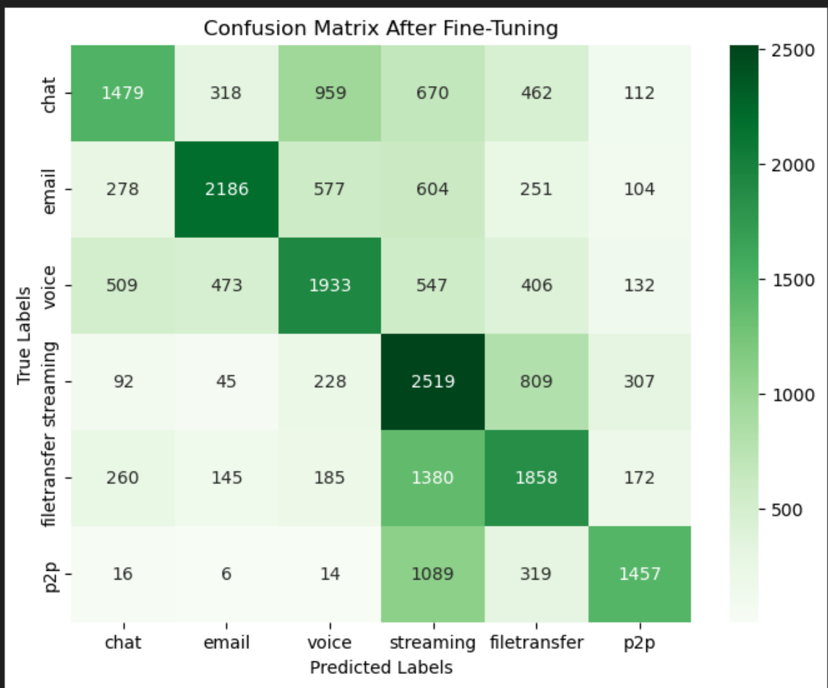
Confusion Matrix After Fine-Tuning
True Labels
Predicted Labels
chat
email
voice
streaming
filetransfer
p2p
1205 559 974 599 578 85
326 1928 773 561 337 75
478 700 1749 513 474 86
87 85 183 2322 1133 190
365 187 308 1167 1859 114
25 7 22 1003 559 1285
2000
1500
1000
500

20kMORTON con softplus + 25 capas descongeladas:

```
1432/1432 [==============================] - 40s 27ms/step - loss: 1.3906 - accuracy: 0.4394
358/358 [==============================] - 8s 19ms/step
```

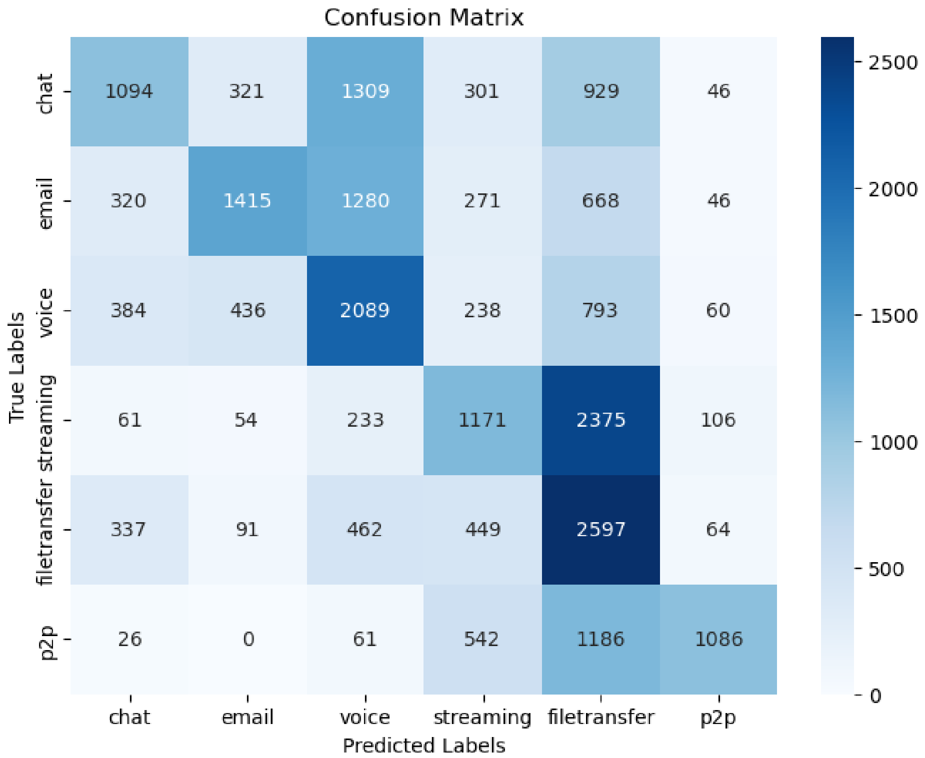


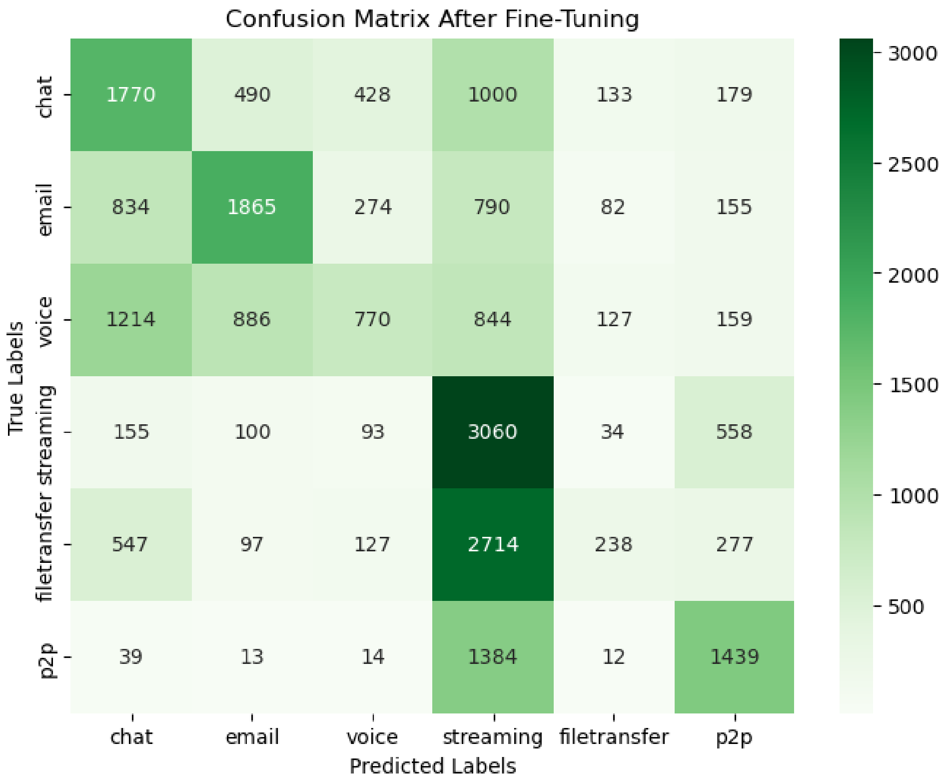

```
Epoch 3/3
1432/1432 [==============================] - 56s 39ms/step - loss: 1.3070 - accuracy: 0.4846 -
358/358 [==============================] - 7s 18ms/step
```



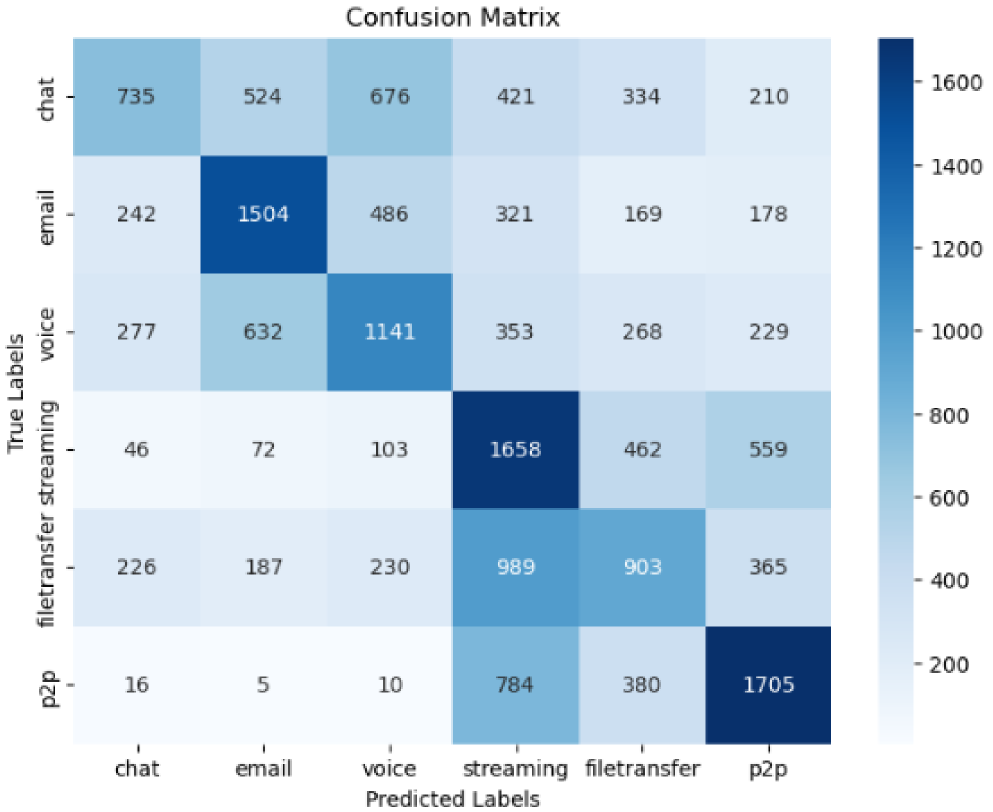

```
358/358 [==============================] - 7s 21ms/step - loss: 1.2870 - accuracy: 0.4992
Test Accuracy after fine-tuning: 49.92%
```

20kMORTON relu softmax:

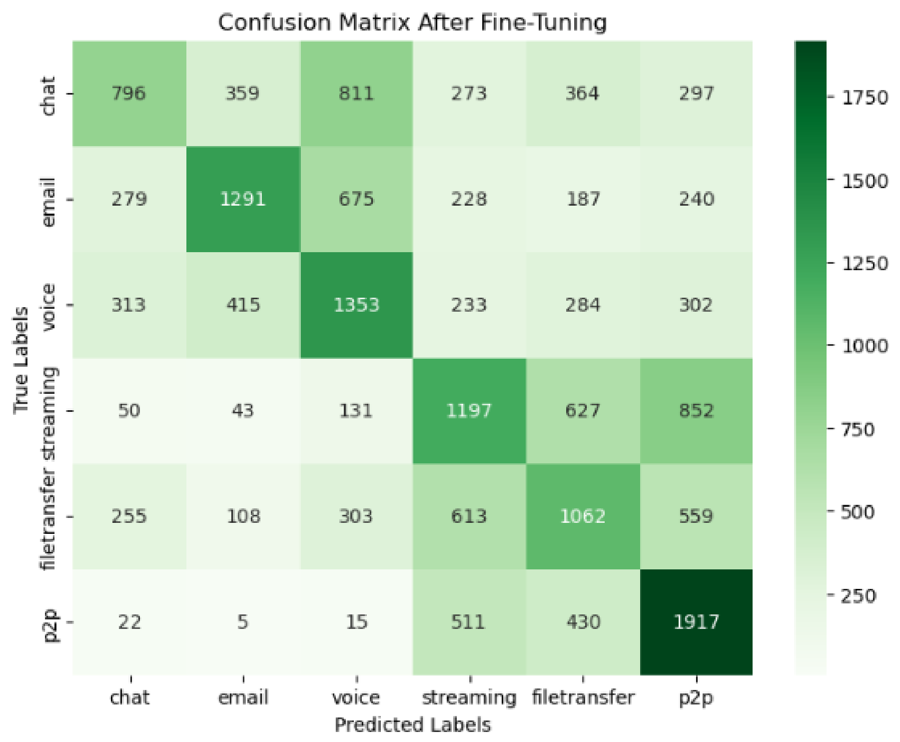




14kMorton:

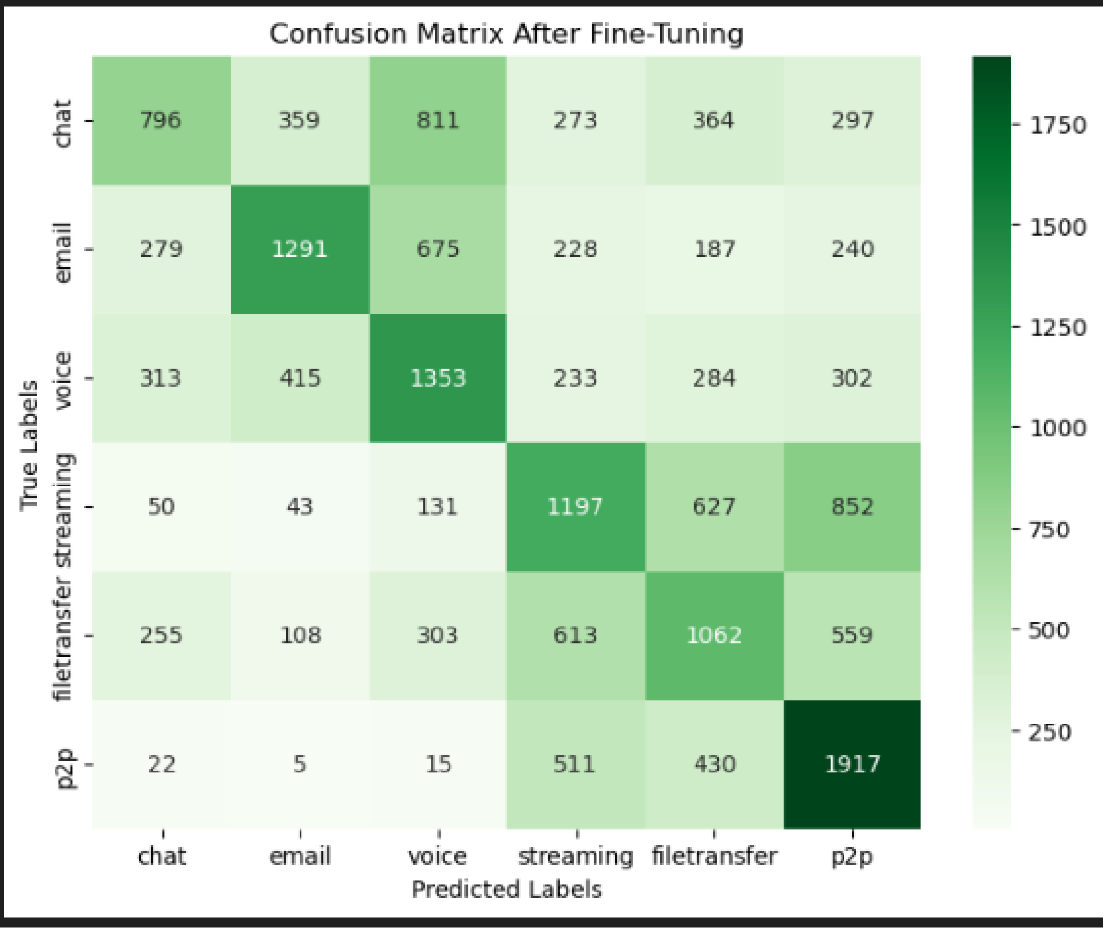



```
Epoch 3/3
1088/1088 [==============================] - 32s 29ms/step - loss: 1.3823
272/272 [==============================] - 7s 21ms/step
```

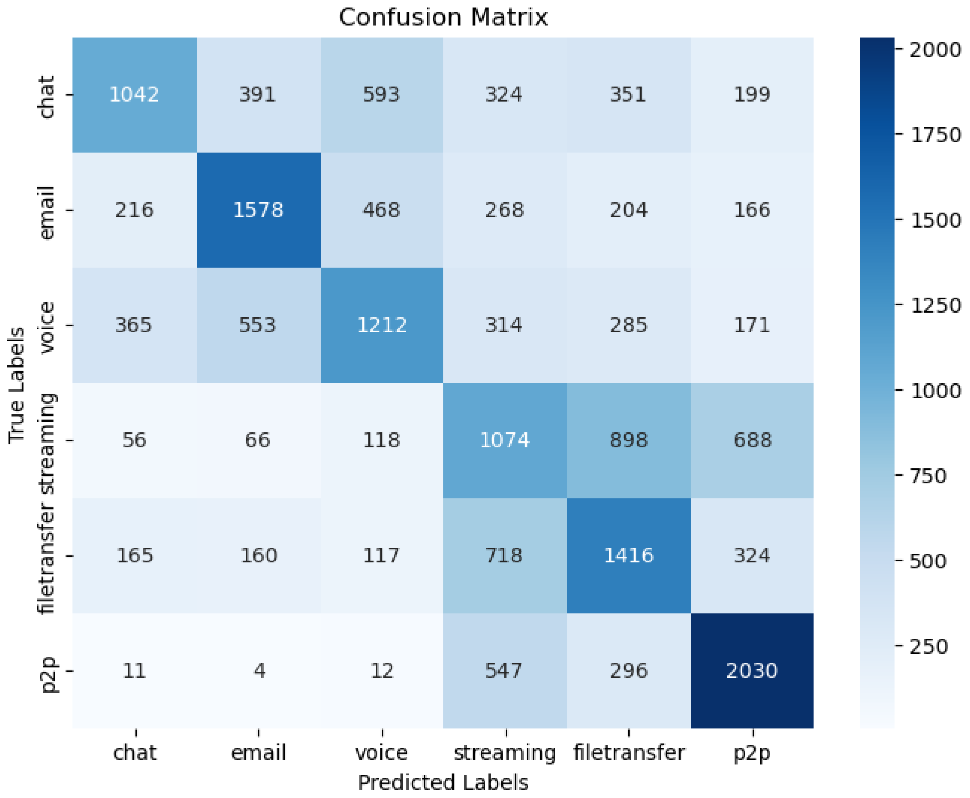


Morton14k 25 capas lr = 0.0001:

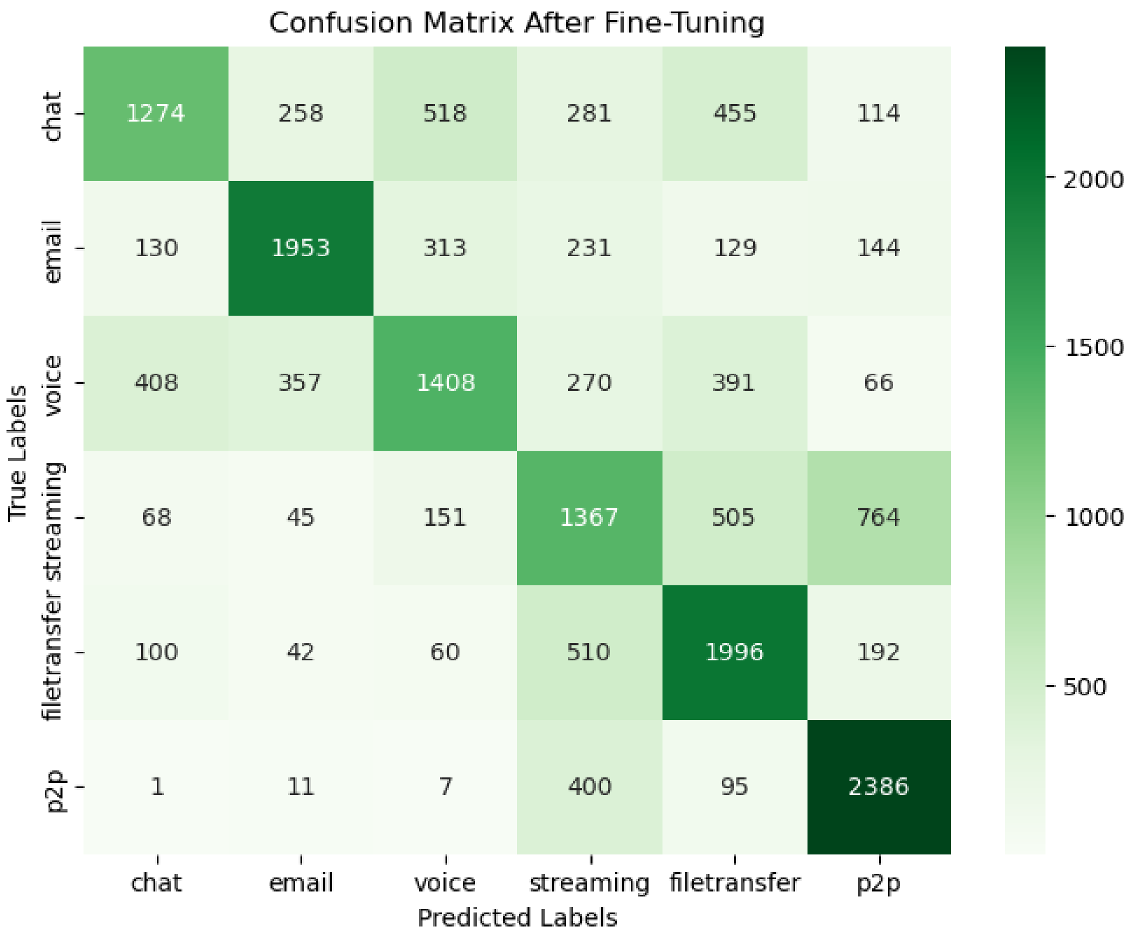
```
Epoch 3/3
1088/1088 [==============================] - 32s 29ms/step - loss: 1.3823
272/272 [==============================] - 7s 21ms/step
```



VGG19:

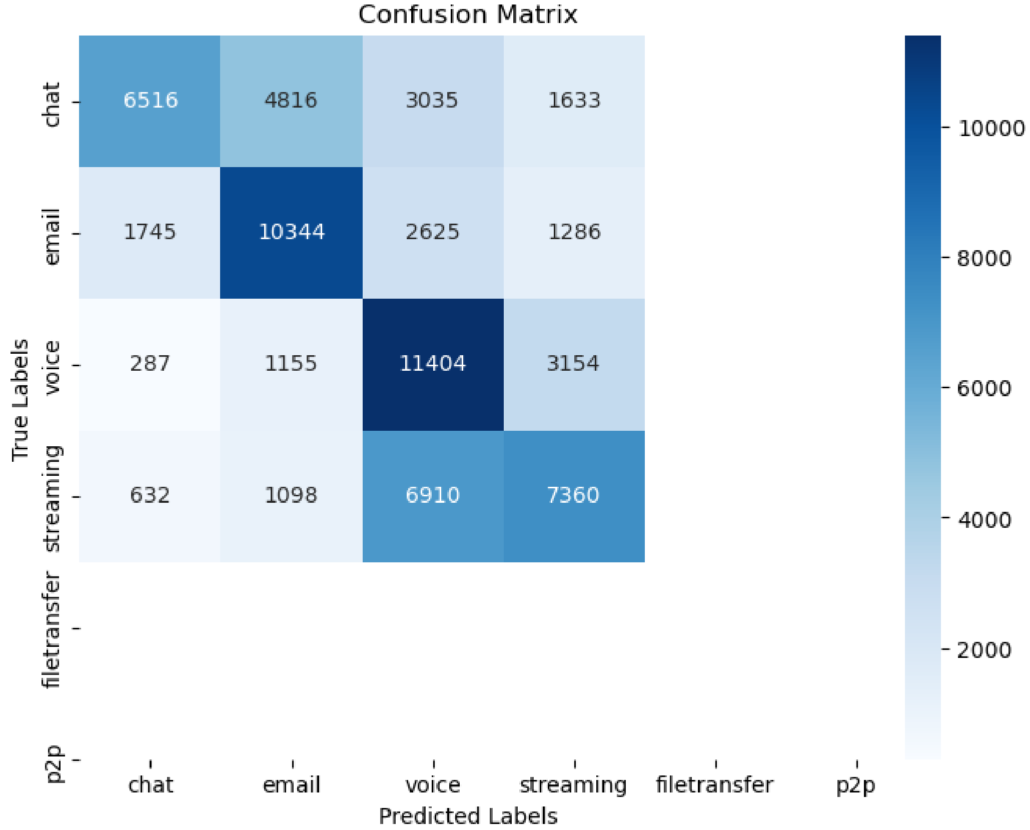


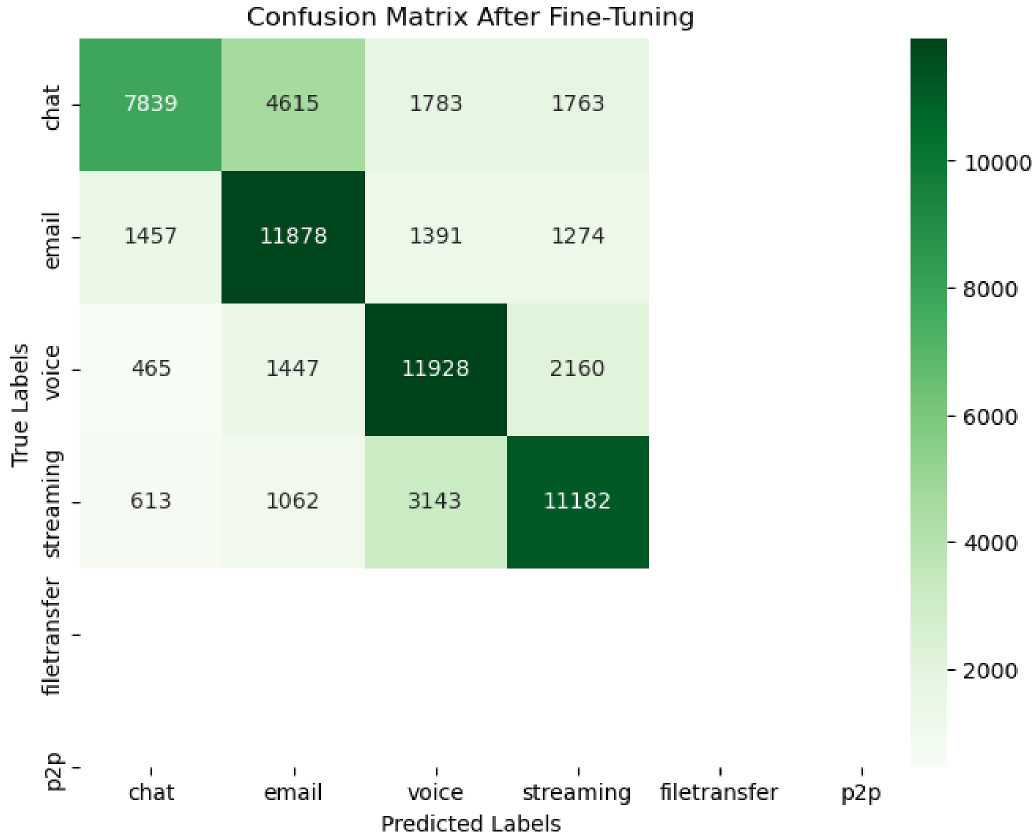
Confusion Matrix
chat
email
voice
streaming
filetransfer
p2p
True Labels
1042 391 593 324 351 199
216 1578 468 268 204 166
365 553 1212 314 285 171
56 66 118 1074 898 688
165 160 117 718 1416 324
11 4 12 547 296 2030
chat
email
voice
streaming
filetransfer
p2p
Predicted Labels
2000
1750
1500
1250
1000
750
500
250



Confusion Matrix After Fine-Tuning
chat
email
voice
streaming
filetransfer
p2p
True Labels
1274 258 518 281 455 114
130 1953 313 231 129 144
408 357 1408 270 391 66
68 45 151 1367 505 764
100 42 60 510 1996 192
1 11 7 400 95 2386
chat
email
voice
streaming
filetransfer
p2p
Predicted Labels
2000
1500
1000
500

VGG19 CON 4 CLASES 80k datos:





DESCARGAMOS EL DATASET NONTOR:

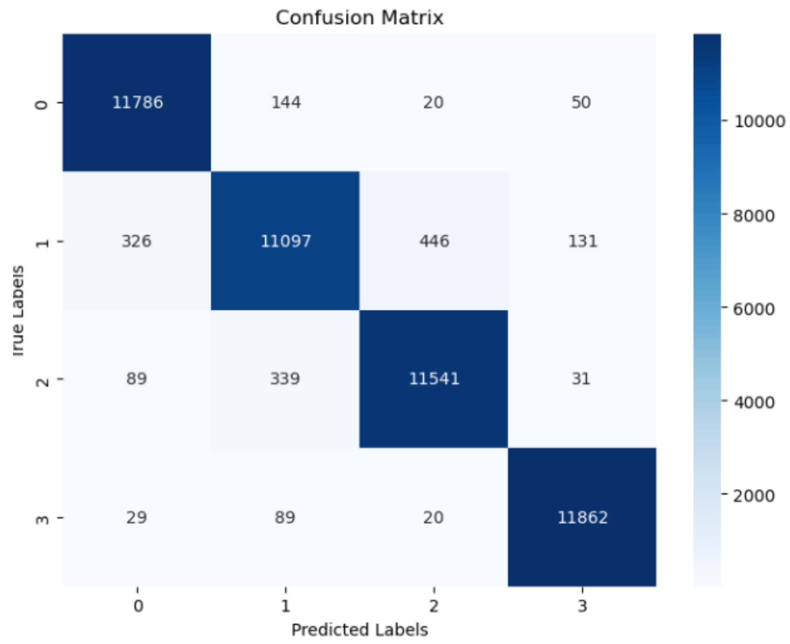
```
Peso total por clase (KB):

label
streaming       576808.12
browsing        511816.63
voice           210925.24
email           191967.80
filetransfer    176125.66
chat              7341.33
Name: size_kb, dtype: float64

Total archivos: 35
```

VS EL DATASET QUE YA TENIAMOS:

```
Peso total por clase (KB):

label
streaming       1420528.12
filetransfer     276920.61
voice             40506.22
chat              25805.89
p2p               19526.14
email              7572.58
Name: size_kb, dtype: float64

Total archivos: 31
```

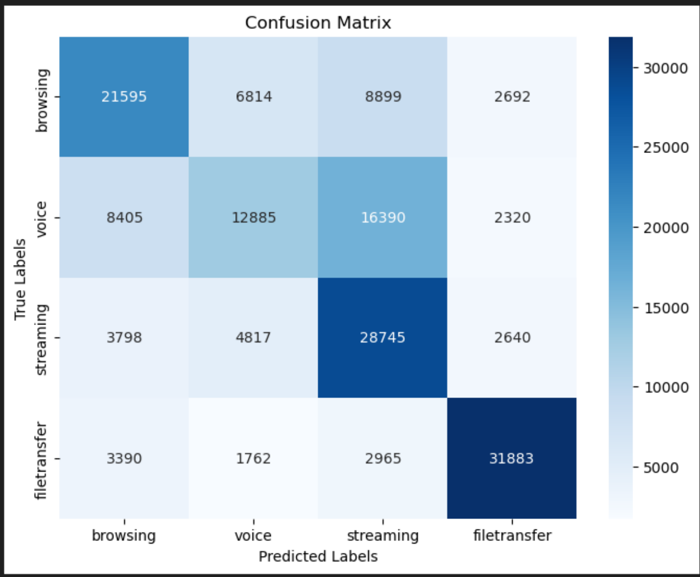
MOVING FORWARD: terminar de ejecutar la transformación de .bin a Morton, y una vez tengamos todas las imágenes, valorar en numero de datos por clase, y tomar una decisión final sobre que aproach tomar, que numero de imágenes por clase, que clases

Con el cambio de reescaling aka hay data leak POR LA CARA

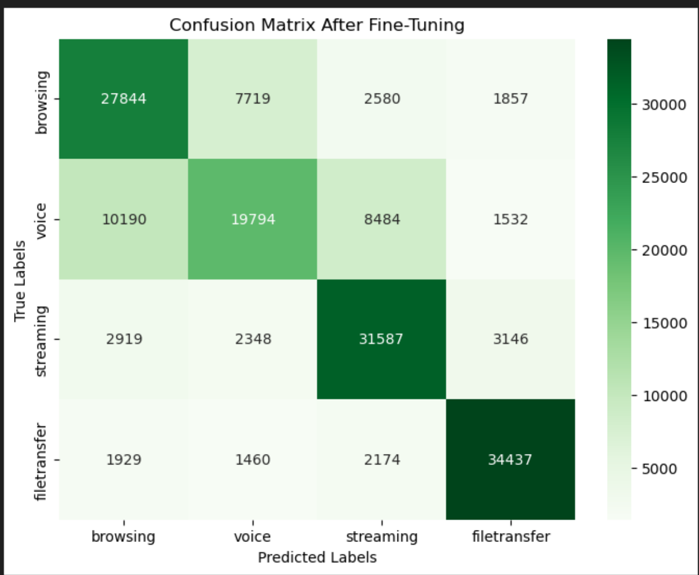


VGG19 con tor 200k:

```
10000/10000 [==============================] - 284s 28ms/step - loss: 0.9364 - accuracy: 0.5917 - val_loss: 0.9343 - val_accuracy: 0.5944
2500/2500 [==============================] - 53s 21ms/step
```

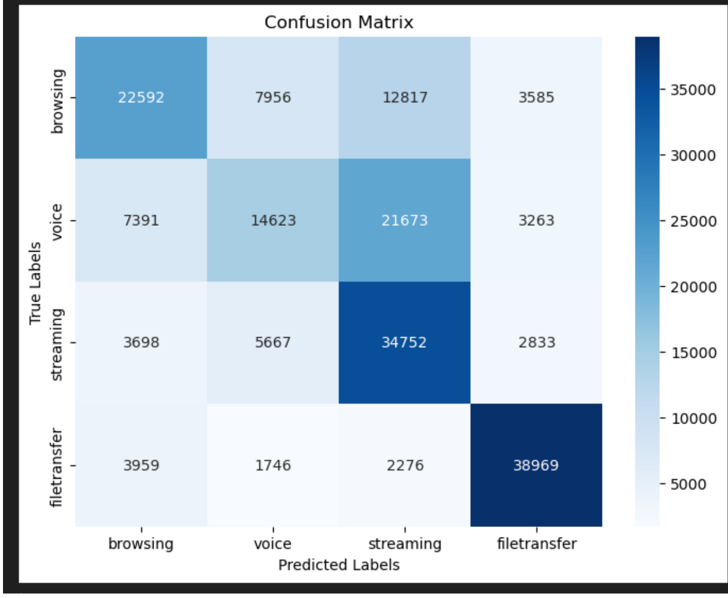



```
Epoch 4/4
10000/10000 [==============================] - 863s 86ms/step - loss: 0.6871 - accuracy: 0.7117 - val_loss: 0.6970 - val_accuracy: 0.7104
2500/2500 [==============================] - 56s 22ms/step - loss: 0.6970 - accuracy: 0.7104
Test Accuracy after fine-tuning: 71.04%
2500/2500 [==============================] - 52s 21ms/step
```

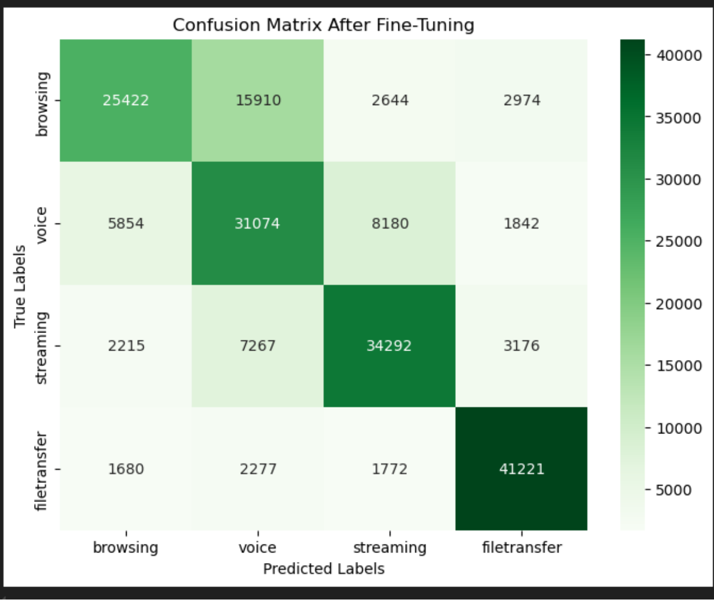


VGG19 con mix de tor/vpn 300k + steps per epoch (1500) + 8 capas unfrozen:

```
Epoch 10/10
1500/1500 [==============================] - 101s 67ms/step - loss: 0.9321 - accuracy: 0.5889 - val_loss: 0.9325 - val_accuracy: 0.5907
2935/2935 [==============================] - 59s 20ms/step
```

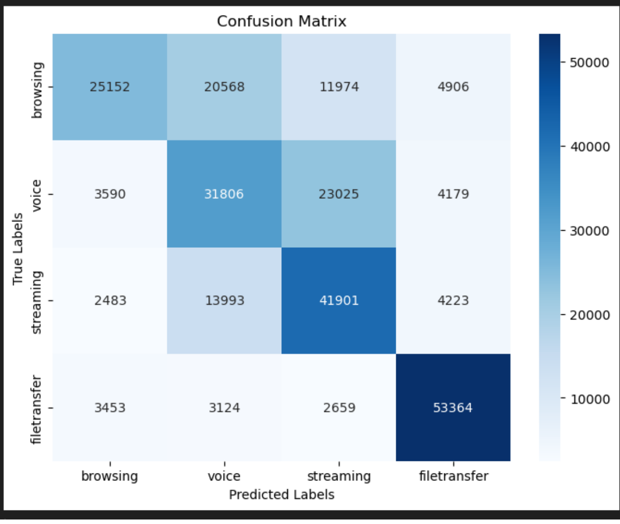



```
Epoch 8/8
1500/1500 [==============================] - 134s 89ms/step - loss: 0.7088 - accuracy: 0.7032 - val_loss: 0.6996 - val_accuracy: 0.7029
2935/2935 [==============================] - 64s 22ms/step - loss: 0.6996 - accuracy: 0.7029
Test Accuracy after fine-tuning: 70.29%
2935/2935 [==============================] - 61s 21ms/step
```

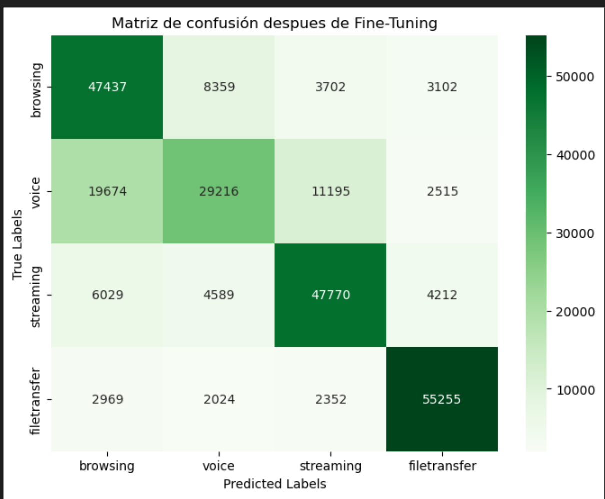


VGG19 con mix de tor/vpn 300k + steps per epoch (500) + lr 0.025 lrFT 0.0005 + Dense ultima capa a 4 no a 6 (cambiando one-hot y dense):

```
Epoch 32/50
148/500 [=======>......................] - ETA: 7s - loss: 0.8920 - accuracy: 0.6040WARNING:tensorflow:Your input ran out of data; interrupting tra
500/500 [==============================] - 87s 175ms/step - loss: 0.8933 - accuracy: 0.6033 - val_loss: 0.8905 - val_accuracy: 0.6079 - lr: 0.0012
3913/3913 [==============================] - 80s 20ms/step
```



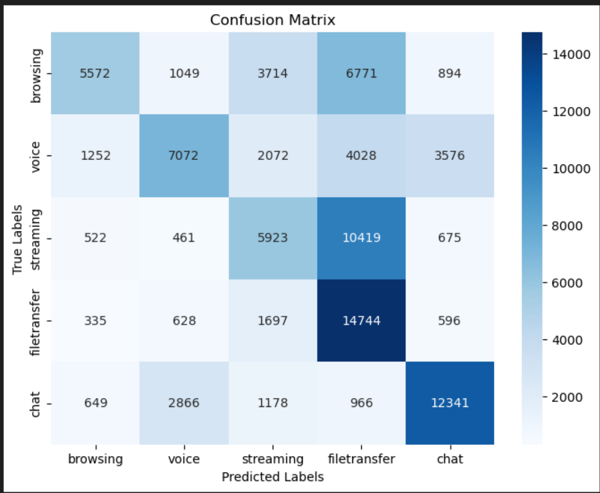

```
Epoch 32/40
150/500 [========>.....................] - ETA: 16s - loss: 0.6764 - accuracy: 0.7142WARNING:tensorflow:Your input ran out of data; interrupting traini
500/500 [==============================] - 91s 182ms/step - loss: 0.6764 - accuracy: 0.7142 - val_loss: 0.6703 - val_accuracy: 0.7176 - lr: 2.5000e-05
3913/3913 [==============================] - 87s 22ms/step - loss: 0.6703 - accuracy: 0.7176
Test Accuracy after fine-tuning: 71.76%
3913/3913 [==============================] - 82s 21ms/step
```



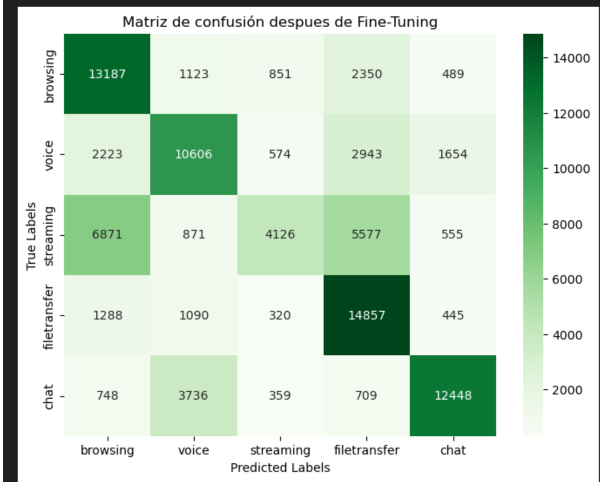

```
Classification Report (After Fine-Tuning):
              precision    recall  f1-score   support

    browsing       0.62      0.76      0.68     62600
       voice       0.66      0.47      0.55     62600
   streaming       0.73      0.76      0.75     62600
filetransfer       0.85      0.88      0.87     62600

    accuracy                           0.72    250400
   macro avg       0.72      0.72      0.71    250400
weighted avg       0.72      0.72      0.71    250400

Weighted F1-score (After Fine-Tuning): 0.7113
Weighted Precision (After Fine-Tuning): 0.7170
Weighted Recall (After Fine-Tuning): 0.7176
```

80k Morton 5 clases con todo el resto de cosas como la anterior:

```
Epoch 8/50
 25/800 [..............................] - ETA: 17s - loss: 1.1937 - accuracy: 0.5069WARNING:tensorflow:Your input ran out of data; interrupting tra
800/800 [==============================] - 30s 38ms/step - loss: 1.1937 - accuracy: 0.5069 - val_loss: 1.1686 - val_accuracy: 0.5072 - lr: 0.0015
1407/1407 [==============================] - 29s 20ms/step
```




```
Epoch 12/40
125/500 [======>.......................] - ETA: 17s - loss: 0.9546 - accuracy: 0.6181WARNING:tensorflow:Your input ran out of data; interrupting train
500/500 [==============================] - 35s 71ms/step - loss: 0.9546 - accuracy: 0.6181 - val_loss: 0.9499 - val_accuracy: 0.6136 - lr: 5.0000e-05
1407/1407 [==============================] - 30s 21ms/step - loss: 0.9499 - accuracy: 0.6136
Test Accuracy after fine-tuning: 61.36%
1407/1407 [==============================] - 28s 20ms/step
```




```
Classification Report (After Fine-Tuning):
              precision    recall  f1-score   support

    browsing       0.54      0.73      0.62     18000
       voice       0.61      0.59      0.60     18000
   streaming       0.66      0.23      0.34     18000
filetransfer       0.56      0.83      0.67     18000
        chat       0.80      0.69      0.74     18000

    accuracy                           0.61     90000
   macro avg       0.63      0.61      0.59     90000
weighted avg       0.63      0.61      0.59     90000

Weighted F1-score (After Fine-Tuning): 0.5945
Weighted Precision (After Fine-Tuning): 0.6347
Weighted Recall (After Fine-Tuning): 0.6136
```

Preparación datos: En normalVPN

0	streaming	856883
1	filetransfer	234224
2	voice	65606
3	chat	46063
4	p2p	14502
5	email	12628

En normalVPN2

	Cantidad	count
0	streaming	312560
1	voice	1553

En normalNoVPN2

	Cantidad	count
0	filetransfer	378941
1	streaming	214169
2	voice	79821
3	chat	34025
4	email	9664

En normalNoVPN

	Cantidad	count
0	streaming	337391
1	voice	708

En nontor/normal

0	voice	613813
1	browsing	419343
2	email	51865
3	chat	8861
4	filetransfer	2161

En nontor/normal2

0	streaming	317992
1	browsing	52383
2	filetransfer	42208
3	chat	2114

TOTAL: streaming **2.038.995**, voice **760.793**, filetransfer: **709.399**, browsing **471.726**, email **74.157**, chat **91.063**, p2p **14,502**

Gameplan: Solo c ojer de no vpn y no tor, y tenemos el mínimo de datos que es la carpeta de browsing, así no mezclamos con datos de VPN lo cual haría que se rayase. Para nontor/normal cojemos 420k x clase

Al final solo pasamos los datos no vpn y nontor